Web APIs

Software Architecture

Brae Webb

April 20, 2026

Goals

- Review existing networking knowledge.

Goals

- Review existing networking knowledge.
- Understand *URLs*.

Goals

- Review existing networking knowledge.
- Understand *URLs*.
- Understand *HTTP* protocol and methods.

Goals

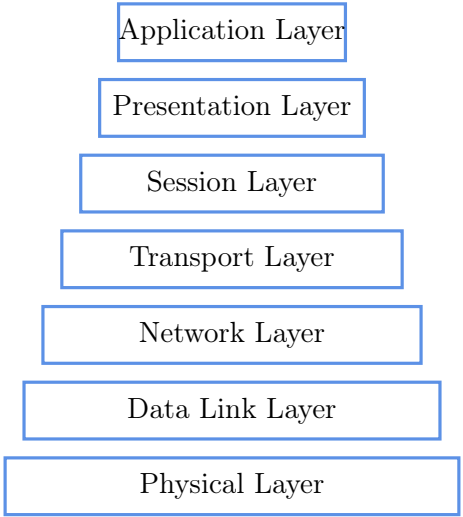
- Review existing networking knowledge.
- Understand *URLs*.
- Understand *HTTP* protocol and methods.
- Understand *RESTful* APIs.

Goals

- Review existing networking knowledge.
- Understand *URLs*.
- Understand *HTTP* protocol and methods.
- Understand *RESTful* APIs.
- *Build* a basic RESTful API.

§ Networking

OSI Model


Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer

OSI Model

Application Layer

Presentation Layer

Session Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer

OSI Model

Application Layer

Presentation Layer

Session Layer

Transport Layer — TCP/UDP (CSSE2310)

Network Layer

Data Link Layer

Physical Layer

TCP/UDP

Low-level with *minimal abstraction*.

TCP/UDP

Impractical for building web APIs.

OSI Model

Application Layer

Presentation Layer

Session Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer

OSI Model

Application Layer — HTTP/HTTPS (CSSE6400)

Presentation Layer

Session Layer

Transport Layer

Network Layer

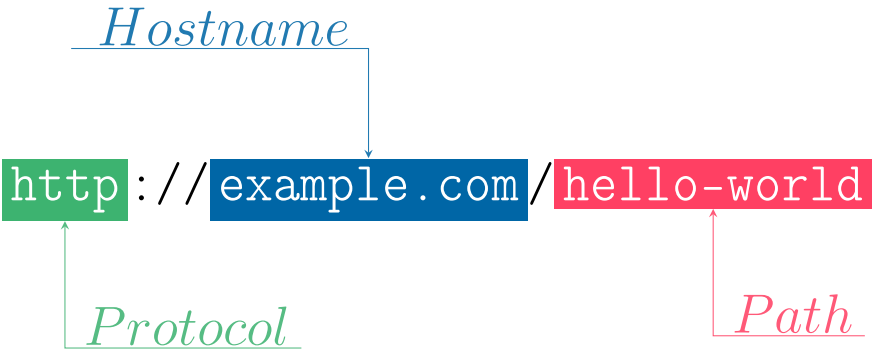
Data Link Layer

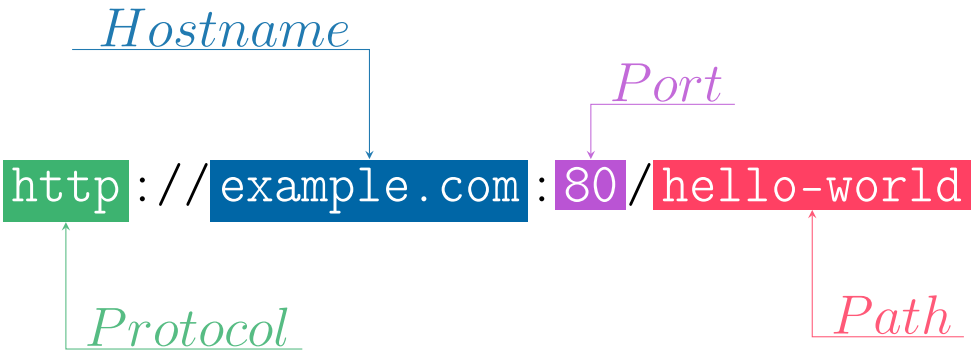
Physical Layer

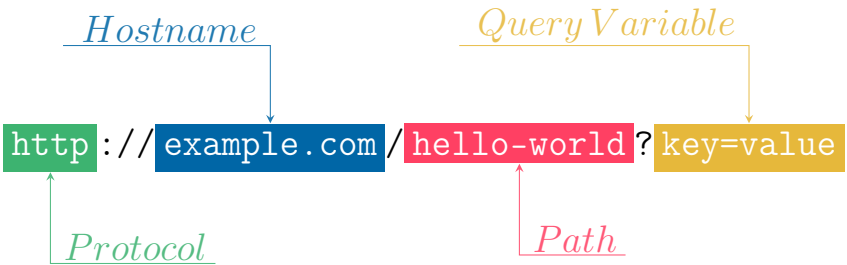
§ URLs

The anatomy of

URLs


Hostname
http://example.com/hello-world
Protocol
Path


Hostname
Port
http://example.com:80/hello-world
Protocol
Path


Hostname
Query Variable
http://example.com/hello-world?key=value
Protocol
Path

§ HTTP

HTTP

A *request-response* abstraction for networking.

HTTP Request

URL An endpoint to send request to.

Method Described later.

Headers Specify type of data, e.g. JSON, HTML, etc.

Body Optional extra data to include.

HTTP Response

Status Code	A number between 100 and 599 giving details about the response.
Headers	Specify type of response data, e.g. JSON, HTML, etc.
Body	Content of the response.

Status Codes

200s Indicate the request was *successful*, 200 is most common.

300s *Redirects* the client to another location.

400s Indicates that the *request was wrong*

e.g. 404 meaning that the request was for something that doesn't exist.

500s Indicates that the *server had a problem* fulfilling the request.

Types of HTTP communication

HTTP Methods

HTTP Methods

GET *Query* for information.

HTTP Methods

GET *Query* for information.

POST *Create* resource.

HTTP Methods

GET *Query* for information.

POST *Create* resource.

PUT *Update* resource.

HTTP Methods

GET *Query* for information.

POST *Create* resource.

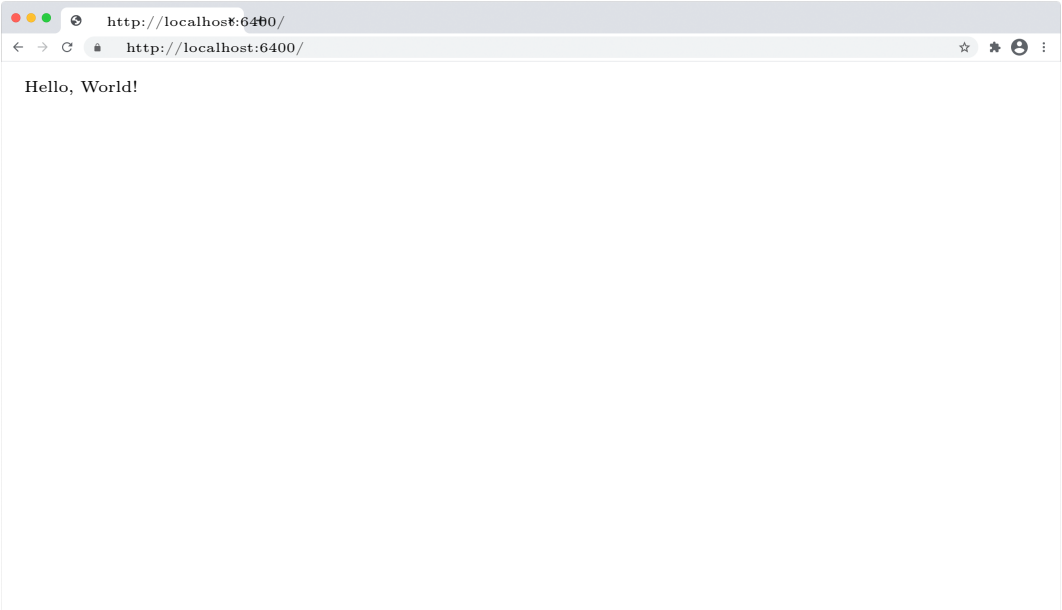
PUT *Update* resource.

DELETE *Delete* resource.

§ API Examples

```
» cat app.py

from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello_world():
    return "Hello, World!"

if __name__ == "__main__":
    app.run(port=6400)
```

Result



```
» cat app.js

const express = require('express')
const app = express()
const port = 6400

app.get('/', (req, res) => {
    res.send('Hello, World!')
})

app.listen(port, () => {
    console.log(`Example app listening on port ${port}`)
})
```

```
» cat app.py

from flask import Flask

app = Flask(__name__)

@app.route("/health")
def hello_world():
    return {"status": "okay!"}

if __name__ == "__main__":
    app.run(port=6400)
```

Result

http://localhost:6400/

http://localhost:6400/

http://localhost:6400/

{"status": "okay!"}

```
» cat app.js

const express = require('express')
const app = express()
const port = 6400

app.get('/', (req, res) => {
    res.send({"status": "okay!"})
})

app.listen(port, () => {
    console.log(`Example app listening on port ${port}`)
})
```

```
» cat app.py

from flask import Flask
from flask import request

app = Flask(__name__)

@app.route("/echo", methods=["POST"])
def hello_world():
    return request.json.say

if __name__ == "__main__":
    app.run(port=6400)
```

```
>>> curl -X POST \
-H "Accept: application/json" \
-H "Content-Type: application/json" \
"http://localhost:6400" \
-d '{
    "say" : "Hello, World",
}'
Hello, World
```

```
» cat app.js

const express = require('express')
const app = express()
const port = 6400

app.post('/', express.json(), (req, res) => {
    res.send(req.body.say)
})

app.listen(port, () => {
    console.log(`Example app listening on port ${port}`)
})
```